EHRA AI checklist — completed for: Estimating atrial fibrillation burden as a per-patient quantification problem with distribution-free uncertainty intervals

Checklist from: Svennberg E, Han JK, et al. State of the Art of Artificial Intelligence in Clinical Electrophysiology in 2025. Europace 2025;27(5):euaf071 (Table 2). Item descriptions are abbreviated; the right-hand column states where each item is addressed in the manuscript. Page numbers to be added by the authors after final typesetting of the submission file.

Reporting standard stated in Methods: TRIPOD+AI and the EHRA AI checklist.

Item	Section/topic	Requirement (abbreviated)	Where addressed
i	Title	Identify the study as using AI/machine learning	Title ("quantification", "distribution-free"); Abstract Methods name the machine-learning detectors and the hidden Markov quantifier. Authors may wish to add "artificial intelligence" to the title.
1	Intended clinical use	Intended use, place in workflow, objective	Introduction (last paragraph); Discussion – Clinical implications: post-processing layer for automated burden from Holter/patch/wearable RR series.
2	Clinical benefit	Added benefit vs standard care/gold standard	Introduction; Results – Burden estimation; Discussion – Clinical implications and Limitations (benefit on decisions not tested).
3	Data collection	How data were collected	Methods – Data sources; Supplementary Table S1; original database publications cited.
4	Source of data	Study design / source of input data	Methods – Study design and Data sources (retrospective, public PhysioNet databases).
5	Development data set	Describe the training data set	Methods – Data sources; Table 1; Supplementary Table S1.
6	Participants	Participants and eligibility criteria	Methods – Data sources and Reference standard (exclusions: invalid RR, unlabelled rhythm, duplicate recording); Table 1. Individual-level demographics not available for most public databases (limitation).
7	Comparator / gold standard	Definition and limitations of ground truth	Methods – Reference standard (database rhythm annotations; AFL counted as non-AF with AF+AFL sensitivity analysis); Discussion – Limitations (annotation quality).
8	Testing data set	Testing set and split definition	Methods – Data partitioning (patient-level stratified splits, five seeds) and external cohorts never used for training/calibration.
9	Sample size	How study size was arrived at	All available annotated patients in each database; Icentia11k random 800-patient subset (computational constraint) – Methods – Data sources; no formal sample-size calculation (limitation).
10	Outcome	Standardised, reproducible, clinically relevant outcome	Methods – Endpoints: per-patient absolute AF-burden error (pp); secondary agreement, threshold classification, coverage.
11	Data type	Data type incl. pre-processing	Methods – Reference standard and units (RR intervals from beat annotations, 0.2–3.0 s, 60-beat windows); Supplementary S2–S3.
12	Data preparation / leakage	Input handling, augmentation, leakage prevention	Methods – Data partitioning and leakage control; Supplementary S2 (automated duplicate audit; per-window features; calibration sets disjoint). No augmentation used.
13	Balanced groups	How/if groups were balanced	Splits stratified by database and AF presence (Methods); no class re-weighting or resampling of windows.
14	Data issues	Poor-quality/noisy/missing/duplicate data	Methods – Reference standard (implausible RR excluded); Supplementary S2 (missing CPSC2021 patient-51 files; duplicate SHDB-AF recording removed; unlabelled Icentia11k segments excluded); robustness to simulated R-peak errors (Results; Figure 5).

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15	Feature engineering	Feature selection and who extracted features	Supplementary S3 (16 RR features computed automatically by the authors' code; no data-driven feature selection).
16	Legal framework	Regulatory approval status	Research software, not approved as a medical device (to be stated by authors in cover letter/Methods if required).
17	Explainability	Patient-level / global / local explainability	The burden layer is a transparent two-state Markov model with interpretable per-patient transition probabilities; posterior burden and interval are reported per patient (Methods; Supplementary S4). Detectors use interpretable RR features (Classical-LR, LightGBM) or a neural network (RRNet).
18	Ethical approval	Ethics information	Methods – Ethics (public de-identified data; original approvals in cited publications).
19	Fairness	Inclusion of relevant groups	External cohorts from Japan, China and Canada; sex and age subgroups in SHDB-AF (Supplementary Table S12); limited demographic metadata elsewhere (limitation).
20	Data availability / code sharing	Public data and code	Data availability statement: all databases public; full code and per-patient results publicly available.
21	Trial registration	Registration if a trial	Not applicable (retrospective diagnostic-accuracy study); analysis plan pre-registered in version control (Supplementary S1).
22	Participants (results)	Baseline characteristics of internal and external data	Results – Cohorts; Table 1; Supplementary Figure S1.
23	Training performance	Results in training data	RRNet learning curves (Supplementary Figure S4); calibration-B (in-distribution) burden and interval results (Supplementary Table S5).
24	Internal validation	Results in testing data	Development test set results (Table 2, Table 5, Supplementary Table S2).
25	External validation	Results in external data	SHDB-AF, CPSC2021, Icentia11k (Tables 2–5; Figures 2–6).
26	Model performance metrics	Appropriate metrics	Window AUROC/AUPRC/sensitivity/specificity/ECE; burden MAE, bias, LoA, CCC, threshold sensitivity/specificity, interval coverage/width.
27	Performance errors	Error analysis	Results – Error analysis; Figure 6; Supplementary Tables S6, S8, S13; Supplementary Figure S7.
28	Comparison with classic statistical methods	What did the model add?	Eight comparators including classical counting, smoothing and statistical quantification estimators (Table 4); classical logistic-regression detector.
29	Generalizability	Discussion of generalisability	Discussion – Interpretation, Comparison with previous work and Limitations.
ii	Conclusion	Supported by the data?	Conclusions restricted to the measured endpoints; effect size described as modest.